

TECHNICAL REPORT: FINAL SUBMISSION

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Project ID	CMPG325-2026-006
Client ID	CLI-006

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1. INTRODUCTION & PROJECT BACKGROUND

This technical report constitutes the formal final submission for the CMPG 325 Computer Networks individual project, consolidating the complete lifecycle from initial client problem analysis through to final network implementation, verification, and client acceptance. The overarching goal of this project was to design, deploy, test, and document a robust, highly secure, and fault-tolerant local area network for **Naledi Specialist Physicians**.

The medical environment requires strict isolation of sensitive medical records, reliable real-time administrative access, secure communications for doctors, and separate, high-priority security handling for closed-circuit television (CCTV) surveillance feeds. To satisfy these operational demands, the network simulation was built in Cisco Packet Tracer using an enterprise-grade hierarchical architecture comprising a Cisco 2911 Router, a Cisco 3560-24PS Core Switch, and two Cisco 2960 Access Switches. Furthermore, Spanning Tree Protocol (STP) was implemented as the core technical feature to maintain network resilience through path redundancy while actively preventing Layer 2 loops.

2. NETWORK DESIGN & PLANNING (MILESTONE 1 CORE)

Naledi Specialist Physicians presented specific operational requirements that shaped the network's logical and physical layout.

2.1 Departmental VLAN Structure

To contain broadcast traffic and enhance security posture between distinct hospital divisions, the network was segmented into the following dedicated Virtual Local Area Networks (VLANs):

- **VLAN 10 (Administration):** Dedicated to administrative staff handling scheduling, billing, and operational management.

- **VLAN 20 (Doctors):** Allocated for physicians requiring secure access to patient diagnoses and high-speed core services.

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- **VLAN 30 (Reception):** Configured for front-desk personnel managing patient intake and initial communications.

- **VLAN 40 (Records):** Designed for secure storage, retrieval, and internal transfer of confidential medical files.

- **VLAN 50 (CCTV):** A specialized security VLAN isolating digital video surveillance feeds away from regular departmental traffic.

- **VLAN 99 (Management):** Reserved for out-of-band management and secure administrative control over network infrastructure devices.

2.2 Architectural Topologies

- **Physical Topology:** Built around a centralized Cisco 2911 Router connected via GigabitEthernet0/0 to a Cisco 3560-24PS Core Switch (FastEthernet0/1). The core switch connects via redundant trunk links to Access-Switch-A and Access-Switch-B, which service end-devices and CCTV cameras.

- **Logical Design:** Integrates inter-VLAN routing at the router-core boundary, IEEE 802.1Q trunk encapsulation across inter-switch links, and explicit port-toVLAN access assignments.

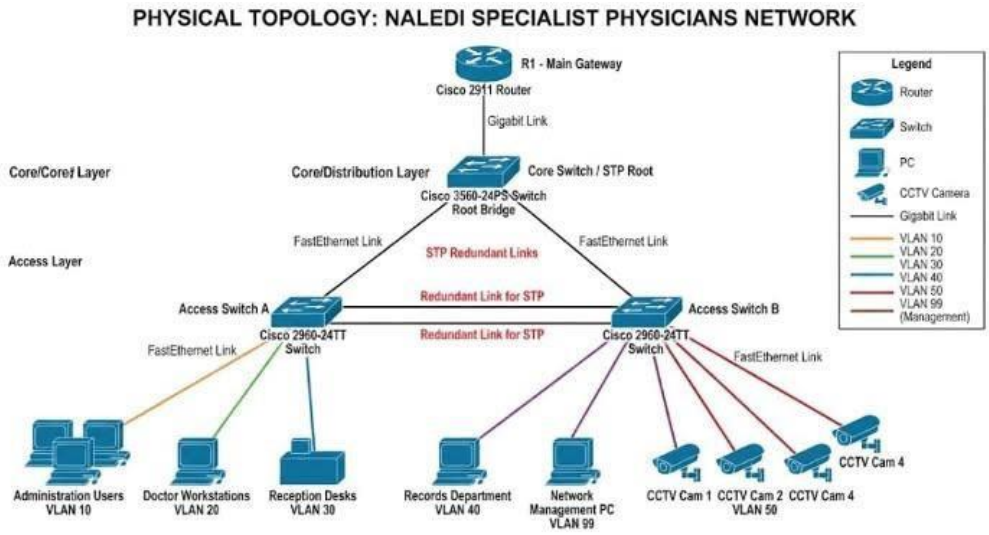


Figure 1: physical topology

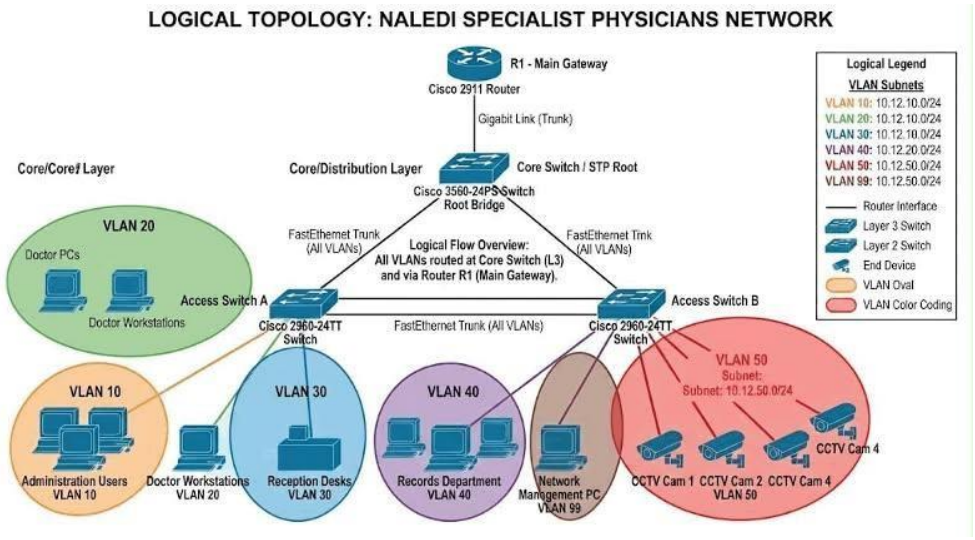


Figure 2: logical topology

3. NETWORK IMPLEMENTATION (MILESTONE 2 CORE)

3.1 Hardware Deployment and Trunking Configuration

The network topology designed in Milestone 1 was fully deployed into Cisco Packet Tracer.

- **Router-to-Core Integration:** Router interface GigabitEthernet0/0 was connected to the Core Switch interface FastEthernet0/1. The interface was brought to an operational up/up state using the no shutdowncommand.

- **Trunk Configuration:** Inter-switch links between the Core Switch and access switches were configured as trunk links using 802.1Q encapsulation. Verification using show interfaces trunk on **Access-Switch-A**and **Access-SwitchB** confirmed that interfaces Fa0/1 and Fa0/2 were actively forwarding all required departmental and security VLANs.



Figure 3: VLAN configuration on Access-Switch-A using `show vlan brief`.

The output confirms that the required VLANs are active on Access-Switch-A, including Administration, Doctors, Reception, Records, CCTV and Management.



Figure 4: VLAN configuration on Access-Switch-B using `show vlan brief`.

4. ASSIGNED FEATURE - SPANNING TREE PROTOCOL (STP)

The assigned technical challenge for this project was the implementation and verification of Spanning Tree Protocol (STP) for loop prevention and root bridge placement.

- **Root Bridge Identification:** The STP root bridge was identified using the bridge MAC address 0004.9AA8.24E4. Verification commands confirmed that the Core Switch and Access-Switch-A participate correctly as core/root elements.

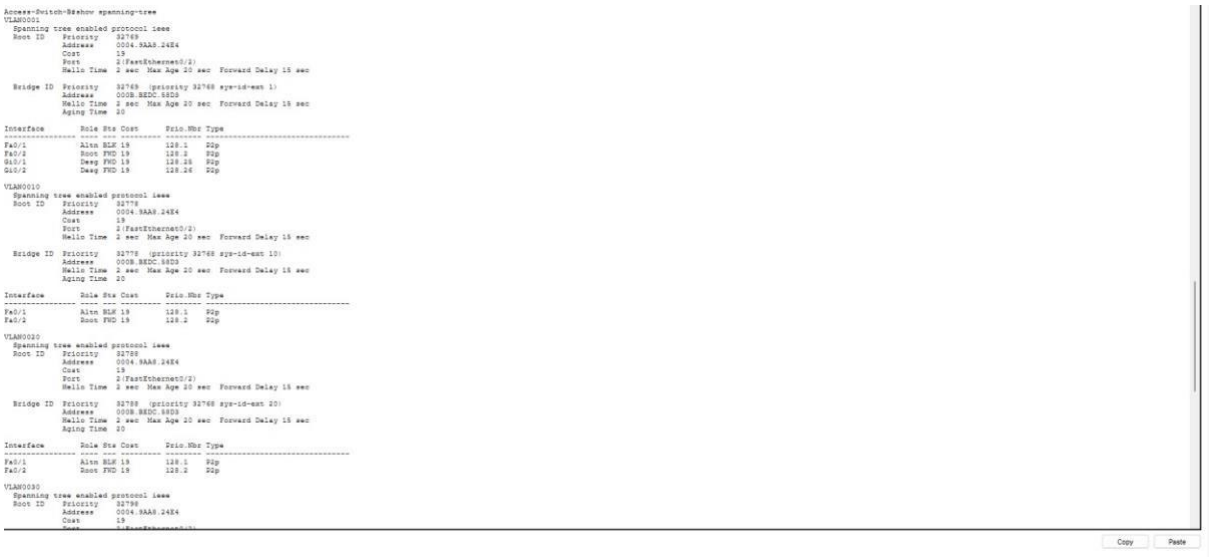
- **Path Selection and Blocking States:** On **Access-Switch-B**, which features dual redundant connections to the core infrastructure, STP successfully evaluated path costs to prevent switching loops:

- Interface Fa0/2 was assigned as **Root Forwarding (Root FWD)**.

- Interface Fa0/1 was placed into an **Alternate Blocking (Altn BLK)** state.

- This demonstrates the fundamental purpose of STP: maintaining high availability through physical redundancy while dynamically blocking redundant active loops during normal operation.

- **CCTV VLAN Isolation:** Four CCTV security cameras were successfully integrated into the dedicated CCTV VLAN (VLAN 50) on Access-Switch-B. Verification outputs confirmed that CCTV ports are correctly bound to VLAN 50 and that STP controls redundant paths for surveillance traffic identically to regular data VLANs.



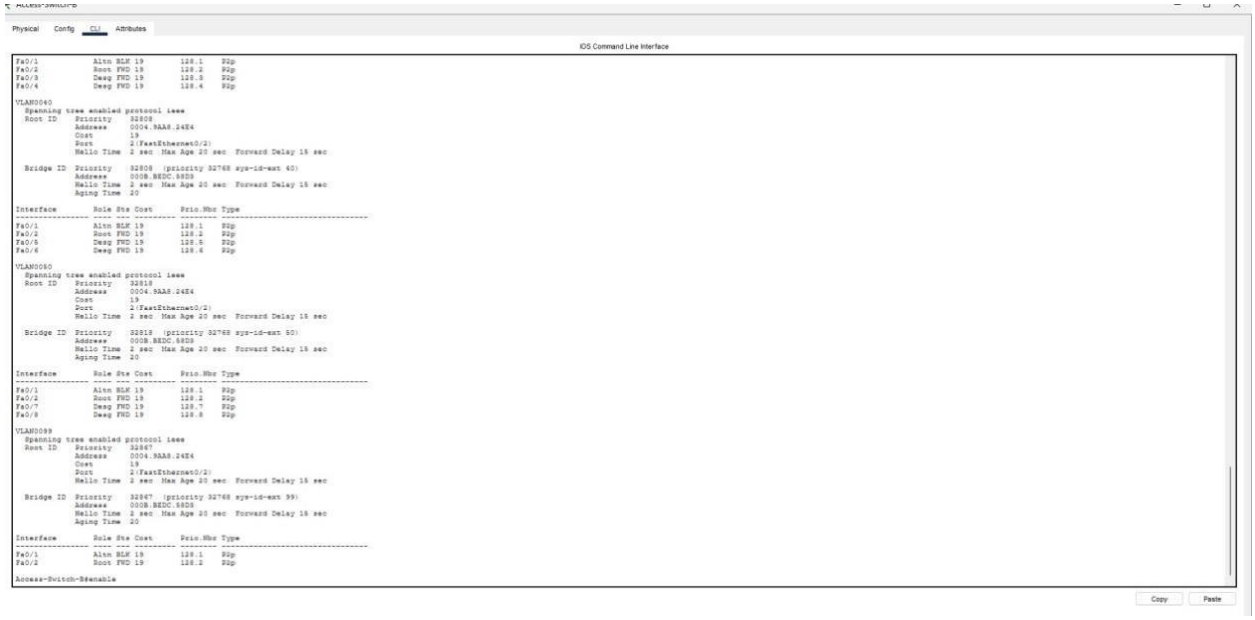
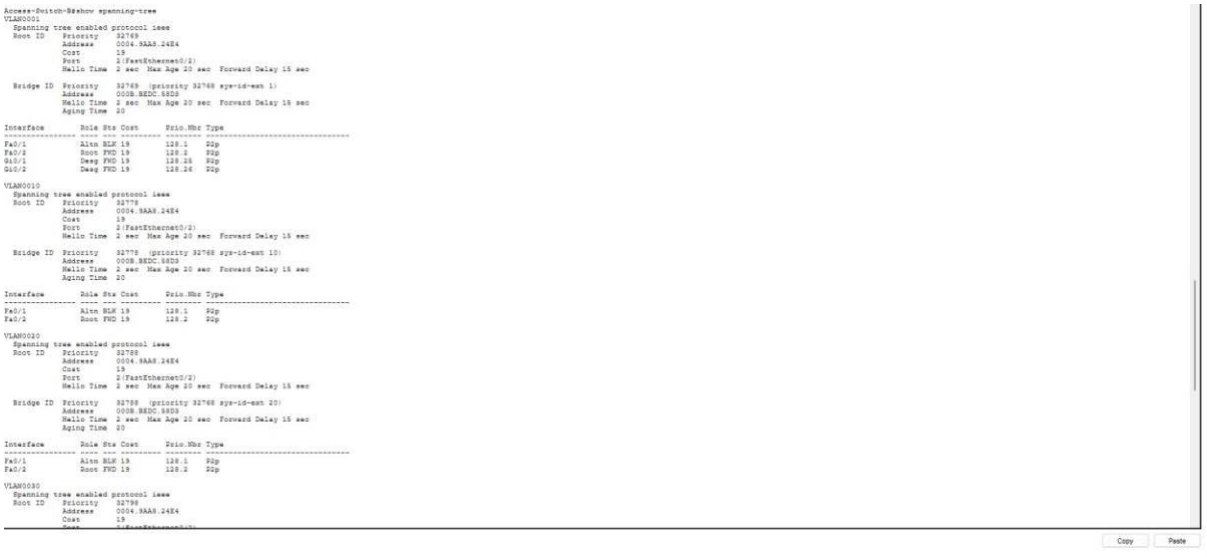


Figure 5: STP verification on Access-Switch-B showing the root and alternate paths.



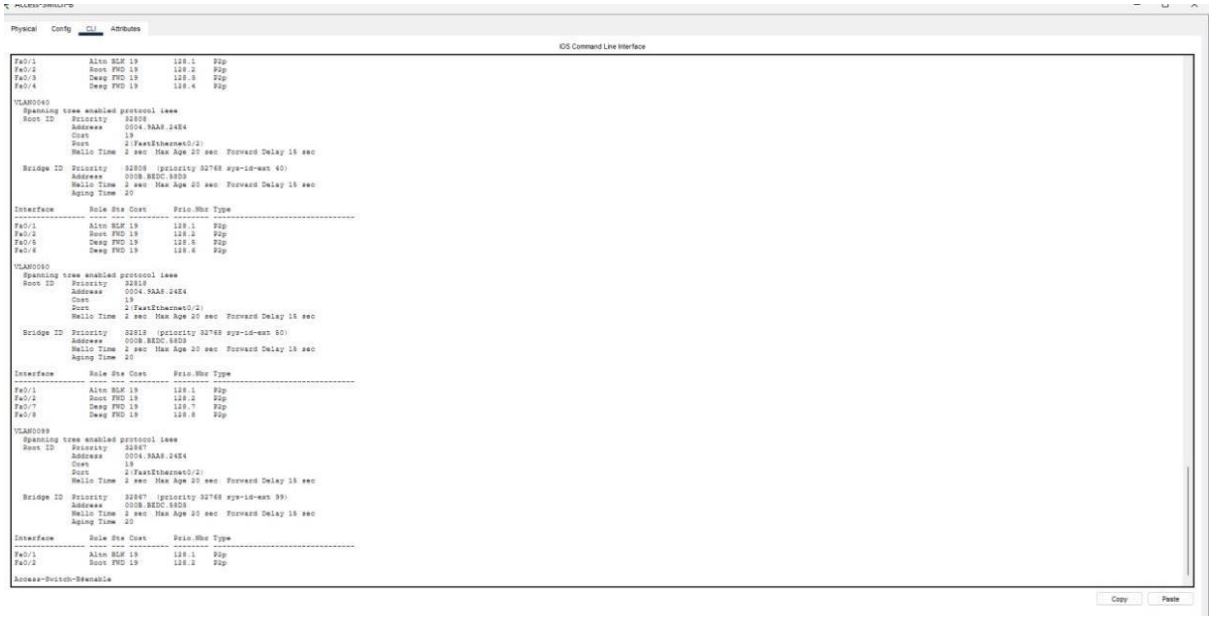


Figure 6:STP verification on Access-Switch-B showing the root and alternate paths

The redundant links between the switches were tested using the `show spanningtree` command.



Figure 7:Final STP verification on Access-Switch-B showing Fa0/2 as Root FWD and Fa0/1 as Altn BLK.

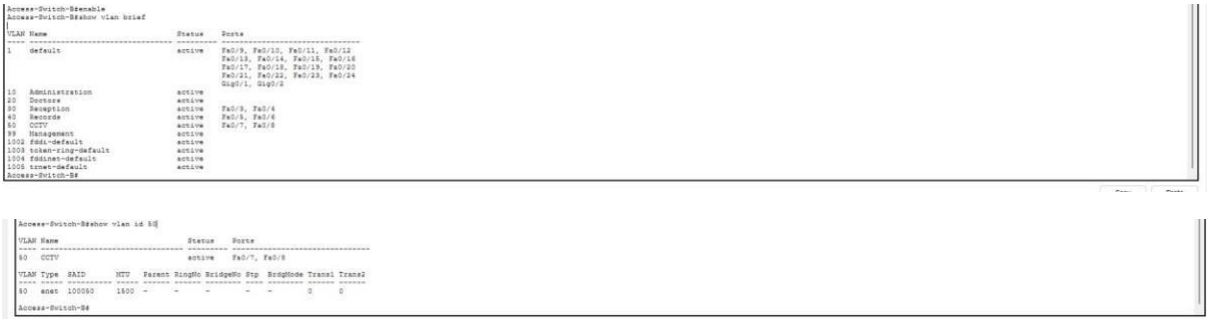


Figure 8: CCTV devices assigned to VLAN 50 on Access-Switch-B.

The output confirms that the CCTV device ports are assigned to VLAN 50.

STP was also verified for VLAN 50.



Figure 9: STP verification for CCTV VLAN 50 on Access-Switch-B.

5. TESTING AND VERIFICATION EVIDENCE

To ensure the network meets all operational requirements, rigorous testing was executed across the simulation environment:

Reception VLAN Connectivity: ICMP ping testing between PCs within VLAN 30 yielded **4 successful replies out of 4 packets with 0% packet loss**, confirming successful intra-VLAN communication.

Records VLAN Connectivity: ICMP ping testing between hosts inside VLAN 40 similarly returned **4 successful replies out of 4 packets with 0% packet loss**.

Device Discovery (CDP): Cisco Discovery Protocol (show cdp neighbors) executed on the Core Switch successfully identified and verified physical connectivity to Access-Switch-A, Access-Switch-B, and Router R1.

Interface Status: Final validation of router interface status confirmed that GigabitEthernet0/0 maintained an operational up/up state. All active device configurations were successfully saved to startup memory.

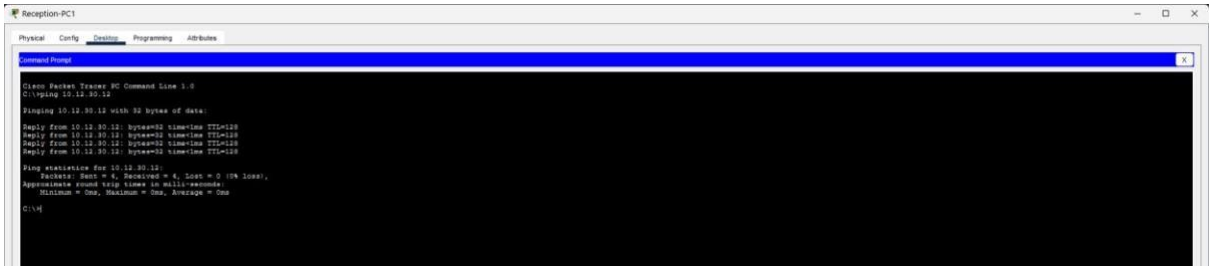


Figure 10: Successful connectivity test between Reception PCs in VLAN 30 showing 4/4 successful replies and 0% packet loss.

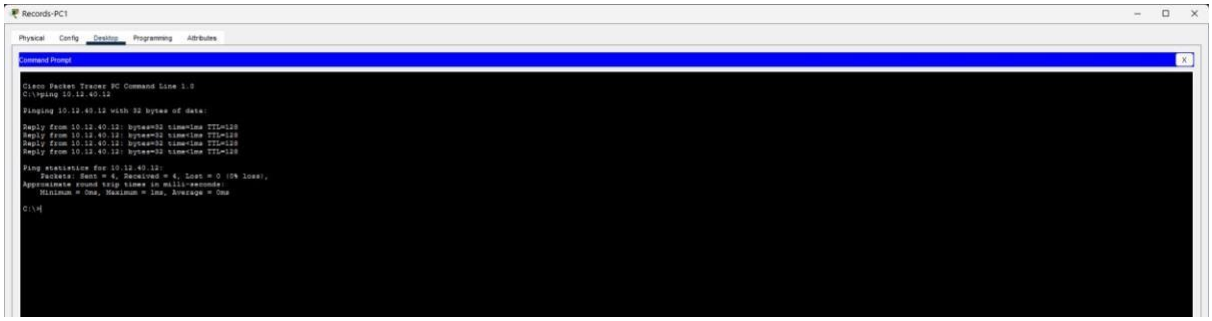


Figure 11: Successful connectivity test between Records PCs in VLAN 40 showing 4/4 successful replies and 0% packet loss.



Figure 12: Core Switch CDP neighbour verification showing the connected access switches and router.

5. GITHUB PORTFOLIO OF EVIDENCE

In compliance with the compulsory evidence streams, a public GitHub repository was maintained throughout the 9-week project lifecycle:

Repository Name: CMPG-325-Naledi-specialist-physicians

Repository Contents: Fully organized directory structures containing Milestone 1 design blueprints, logical topology files, Milestone 2 implementation scripts, verification screenshots, and a comprehensive project README.md file.

6. VIDEO DEMONSTRATION OVERVIEW

A 15–20 minute video demonstration has been recorded and linked via the GitHub portfolio, structured to meet all assessment rubric criteria:

1. **Introduction (0–3 mins):** Student identification, project code (CMPG3252026-006), and client specifications for Naledi Specialist Physicians (CLI-006).

2. **Design Overview (3–7 mins):** Comprehensive walk-through of physical topology diagrams, logical layouts, subnet allocations, and the 6-tier VLAN architecture.

3. **Simulation & Configuration Walkthrough (7–13 mins):** Live demonstration inside Cisco Packet Tracer showing device CLI commands (`show vlan brief`, `show interfaces trunk`, and `show ip interface brief`).

4. **Assigned Feature Analysis - STP (13–18 mins):** Detailed breakdown of `show spanning-tree` outputs on Access-Switch-A and Access-Switch-B, explaining root IDs, port costs, and the operational mechanics of Alternate Blocking (Altn BLK) ports.

5. **Testing & Conclusion (18–20 mins):** Live execution of ping verification tests (demonstrating 0% packet loss) and final project sign-off.

7. CONCLUSION & FINAL REFLECTION

The computer network designed, implemented, and tested for **Naledi Specialist Physicians** fully satisfies all client requirements outlined in project code CMPG3252026-006. By translating theoretical concepts into a functional Cisco Packet Tracer model, the network achieves robust security segmentation, clear departmental isolation, and high availability.

The implementation of Spanning Tree Protocol successfully resolved potential Layer 2 broadcast storms associated with physical switch redundancy. Verification commands confirmed that root bridges operate reliably and alternate paths transition correctly into blocking states. Furthermore, end-to-end testing across VLAN 30 and VLAN 40 verified flawless communication with zero packet loss, while CCTV surveillance traffic was completely isolated into VLAN 50. Through consistent planning, thorough documentation on GitHub, and rigorous testing, this project

delivers a scalable, secure, and professional network ready for immediate client acceptance.